

COURSE CODE: CSE 4204 || COURSE TITLE: MOBILE COMPUTING LAB

WEEK 06 TASK: BACKEND DEVELOPMENT & DATABASE IMPLEMENTATION PROGRESS REPORT

TEAM NAME: CSE4204-8D-T01 || SECTION: 8D

Project Title: NUBTK_COPILOT - AI-Powered University ERP and Career Architect

Instructor: Md. Riaz Mahmud, Assistant Professor, Dept of CSE, NUBTK

Section: 8D || **Team ID:** CSE4204-8D-T01

GitHub Repository: <https://github.com/bashirnubtk/cse4204-8d-t01-nubtk-copilot>

1. Team Information & Core Engineering Roles

Full Member Name	Student ID	System Design & Engineering Responsibilities
Bashir Alam	11220320963	Team Leader. Core backend system architecture, API design, Celery worker workflow orchestration, Open Router API & ML models integration.
Tisha Rani Chakroborty	11220320954	Frontend Developer. UI/UX layout design systems, interactive dashboard wireframes, and responsive views matching via Tailwind CSS.
Fariha Rahman Oishee	11220320967	Database Manager. PostgreSQL schema structure planning, entity relationship mapping, and relational table indexing within Supabase.
Ratul Raha Shuvo	11220320949	Backend Support. Payment gateway integrations sub-system design, transactional report templates, and REST API test suite architecture.

2. Project Objective & Scope

The core objective of this week's development cycle was to transform the theoretical blueprints from the System Design phase into a functional backend system. The key focus areas included:

- Setting up a scalable, decoupled, and asynchronous server infrastructure using Django 5.0.
- Deploying and configuring the live relational database schema on Supabase Cloud Engine.
- Implementing secure authentication protocols including password encryption and JWT session management.
- Developing and deploying initial RESTful API endpoints for user onboarding and core operational workflows.

3. Backend Technology Stack

The infrastructure of **NUBTK_COPILOT** leverages modern industry-standard frameworks to ensure data consistency, security, and parallel task execution capability:

- **Core Framework:** Django 5.0 Core Framework Services
- **Database Engine:** PostgreSQL deployed via Supabase Cloud Engine
- **Asynchronous Engine:** Celery Task Queue for orchestration
- **Authentication Engine:** Django REST Framework SimpleJWT (JSON Web Tokens)
- **Password Security:** Secure PBKDF2 / Bcrypt cryptographic password hashing

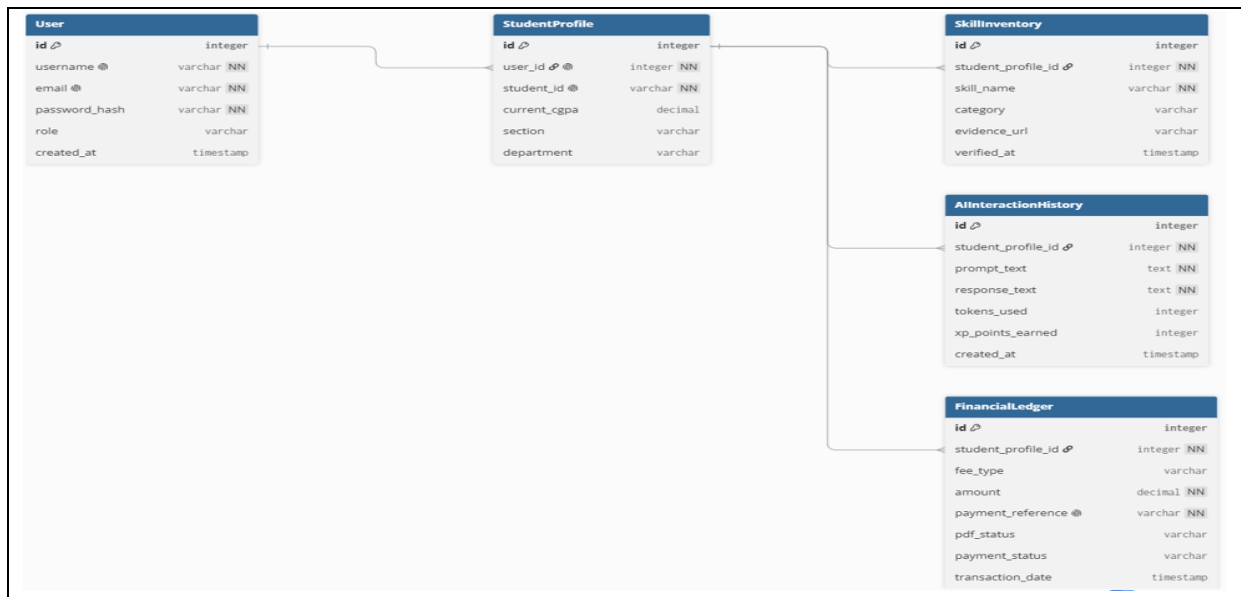
4. Database Architecture & Schema Implementation

The cloud database instance has been configured and tables have been successfully materialized on the Supabase PostgreSQL platform based on the approved ER Diagram structure. The structural layout includes explicit foreign keys, primary keys, and constraint validations:

4.1 Implemented Relational Tables

1. **User Table (Authentication Core):** Manages essential access credentials, roles (Student, Faculty, Admin, Super Admin), and encrypted password blocks.
2. **Student Profile Table:** Keeps track of vital academic data including institutional Student ID registries, GPA scores, and links to user credentials.
3. **Skill Inventory Table:** Acts as an immutable ledger storing practical engineering skills tagged against individual student records.
4. **AI Interaction History Table:** Archives detailed token usage, system chat logs, and contextual context strings from the AI Mentor Hub.
5. **Financial Ledger / Transaction Table:** Logs digital payment gateway reference numbers, enrollment validation tags, and automated accounting workflows.

Database Architecture



User	StudentProfile
id 🔑 integer	id 🔑 integer
username 🔒 varchar NN	user_id 🔑 🔒 integer NN
email 🔒 varchar NN	student_id 🔒 varchar NN
password_hash varchar NN	current_cgpa decimal
role varchar	section varchar
created_at timestamp	department varchar

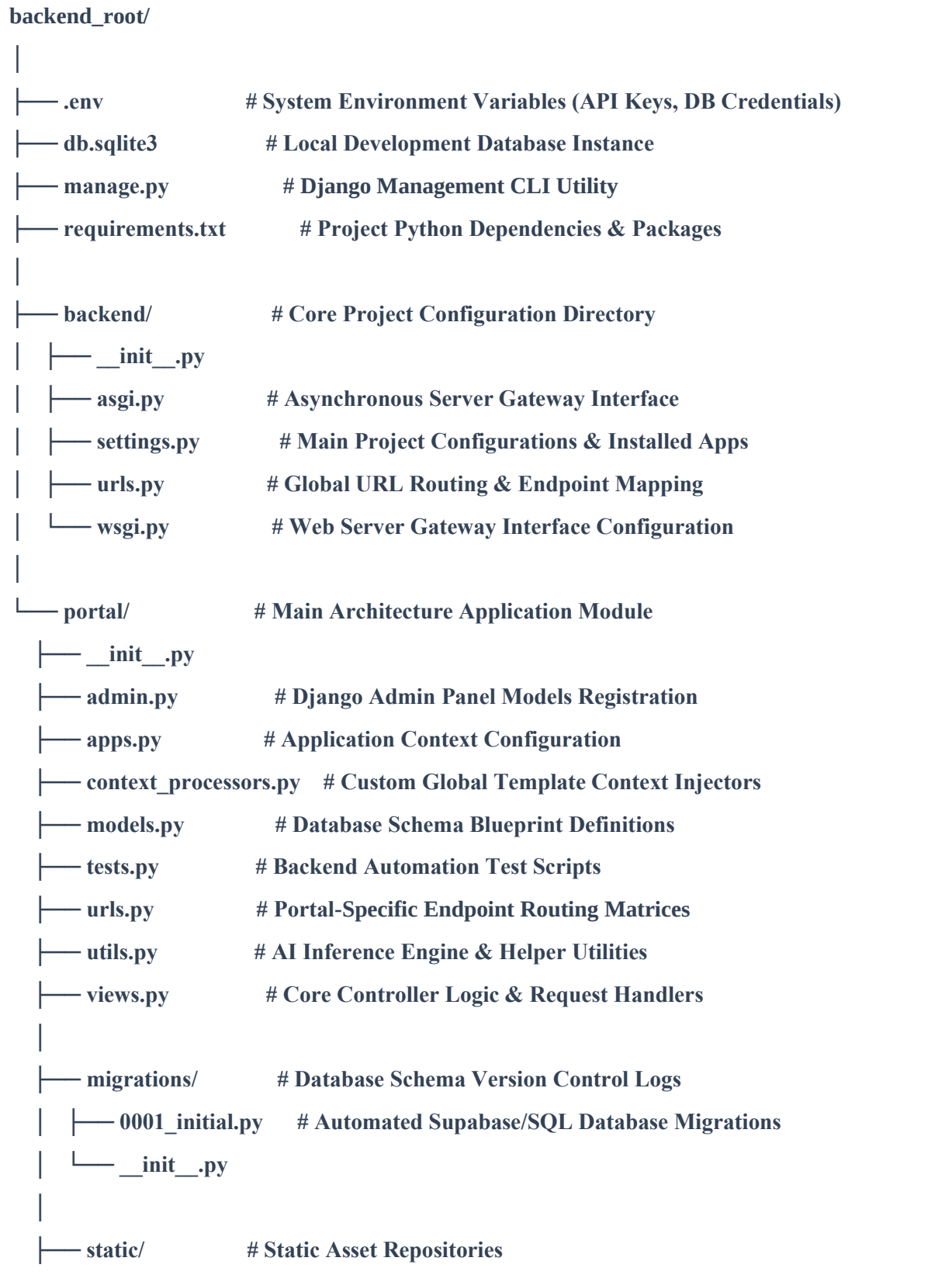
5. Authentication & Authorization Workflow

To safeguard corporate academic identities, a highly secure JWT-based stateless framework was implemented:

- Registration:** The client applications send payloads containing the user credentials to the backend. The application layer interceptor hashes the cleartext passwords using secure PBKDF2 blocks before saving them to the user schema.
- Login Verification:** Upon sending active credentials to the server, the backend system evaluates the cryptographic hash values. If authenticated, the service issues an isolated, time-bound **Access Token** and a long-lived **Refresh Token**.

3. **Route Protection:** Protected API endpoints (e.g., logging skills or launching mentor chats) require a valid Bearer <Token> string embedded directly into the standard incoming HTTP Authorization requests Header.

Backend Folder Structure:



		css/	
		└─ output.css	# Compiled Tailwind CSS Utility Layer
		└─ js/	
		└─ main.js	# Client-Side Interactive Scripts
		└─ templates/	# Server-Side Rendered HTML Layer
		└─ academic_matrix.html	# Semester Course Mapping Interface
		└─ admin_dashboard.html	# Superuser Management Control Center
		└─ base.html	# Main Global Blueprint Layout Shell
		└─ chat.html	# Real-Time AI Copilot Interaction Console
		└─ home.html	# Institutional ERP Entry Landing Page
		└─ login.html	# Secure User Authentication Gateway
		└─ student_profile.html	# Dynamic Skill Ledger & Portfolio View

6. RESTful API Implementation & Endpoint Specification

The backend currently presents a standardized API routing environment. Input parameters pass through data serializers to enforce payload shape validity and emit standard HTTP status codes:

HTTP Verb	API Endpoint Routing	Target Module	Primary System Action / Output Status	Current Status
POST	/api/auth/register/	Authentication	Registers a new user inside the institutional DB repository context.	Completed
POST	/api/auth/login/	Authentication	Validates account credentials and issues secure access JWT.	Completed
POST	/api/skills/log/	Skill Inventory	Appends a verified engineering technical skill node to active student profile.	Completed
POST	/api/ai-mentor/chat/	AI Mentor Hub	Streams context-aware programming guidance generated via Claude 3.	Completed

7. API Testing & Verification Reports

Every deployed API route went through thorough testing protocols using Postman / Thunder Client suites. Validation metrics checked structural behavior on successful transactions as well as fallback performance on erroneous inputs (e.g., duplicate emails, bad validation shapes, missing tokens).

7.1 User Registration Testing (POST /api/auth/register/)

HomePortal LoginAdmission 2026

Application submitted! Your Enrollment ID is NUBTK-2026-1859.

NUBTK Admission Application
Session 2026 - CSE Department Enrollment

Personal Details

Student Full Name *

Enter full legal name

Father's Name *

Father's name

Mother's Name *

Mother's name

Contact Information

Phone Number *

e.g. 017XXXXXXX

Email Address *

student@gmail.com

```
[27/Jun/2026 19:46:57] "GET /portal/admission/apply/ HTTP/1.1" 200 10121
[27/Jun/2026 19:49:51] "POST /portal/admission/apply/ HTTP/1.1" 302 0
[27/Jun/2026 19:49:51] "GET / HTTP/1.1" 200 18015
```

7.2 User Login Testing (POST /api/auth/login/)

HomePortal LoginAdmission 2026

You have been successfully logged out.

NUBTK Pilot Portal
Student Login & Admin Access Gateway

১ রোল সিলেকশন পাইড: স্টুডেন্ট এবং অ্যাডমিন উভয়েই নিজ নিজ রেজিস্টার্ড জিমেইল আখতি দিয়ে প্রবেশ করুন।

Registered Email Address

bashir_alam

Portal Access Password / Digital ID

...

Secure Portal Sign In

```
[27/Jun/2026 21:30:05] "GET / HTTP/1.1" 200 18015
[27/Jun/2026 21:30:07] "GET /portal-admin/login/ HTTP/1.1" 200 8286
[27/Jun/2026 21:30:21] "POST /portal/gateway/login/ HTTP/1.1" 302 0
E:\Projects\Mobile_Computing\Nubtk_Copilot\venv\Lib\site-packages\keras\src\layers\core\dense.py:107: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
[27/Jun/2026 21:30:28] "GET /portal/student/dashboard/ HTTP/1.1" 200 12268
```

1

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed,
GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
[27/Jun/2026 21:30:28] "GET /portal/student/dashboard/ HTTP/1.1" 200 12268
[27/Jun/2026 21:31:54] "GET /student-portal/practical-skills/ HTTP/1.1" 200 9009
```


NUBTK COPILOT

[Home](#)
[Dashboard](#)
[Logout](#)

AI Mentor Core Memory

This live metadata vector is being injected into the LLM system prompt in real-time:

Bashir Alam

75% Ready

Data Science & Machine Learning

3.64

OpenRouter Gateway Active

NUBTK Copilot AI Workspace

Gemini 2.5 Engine

Hello Bashir Alam! I am your AI Mentor.

Ask me anything about your academic track, due bills, or career recommendations.

NUBTK Copilot

নুবটকে হাইব্রিড সার্ভার ক্লাস্টারে অতিরিক্ত ট্রাফিকের কারণে রেসপন্স জেনারেট করা যায়নি। আবার চেষ্টা করুন।

Send

```

ut_shape` / `input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in
he model instead.
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed
GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
[27/Jun/2026 21:30:28] "GET /portal/student/dashboard/ HTTP/1.1" 200 12268
[27/Jun/2026 21:31:54] "GET /student-portal/practical-skills/ HTTP/1.1" 200 9009
[27/Jun/2026 21:32:43] "GET /student-portal/ai-chat-hub/ HTTP/1.1" 200 11067
[27/Jun/2026 21:32:48] "POST /student-portal/ai-chat-hub/ HTTP/1.1" 200 627

```

To strictly conform to professional software engineering standards, continuous integration workflows were practiced. Every distinct code adjustment was organized, committed using descriptive system statements, and published to the central organization repository.

- feat: implemented JWT login system and user authentication routing modules
- db: finalized PostgreSQL schemas and initialized Supabase structural migrations
- fix: resolved structural input data validation errors during student sign-up

bashirubtk / cse4204-8d-t01-nubtk-copilot

Q Type [Z] to search

Code

Issues

Pull requests

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

cse4204-8d-t01-nubtk-copilot

Public

Pin

Watch 0

Fork 0

Star 0

main 6 Branches 0 Tags

Go to file

Add file

Code

bashirubtk Merge pull request #10 from bashirubtk/dev 29d5f44 · 2 weeks ago 25 Commits

documentation Add files via upload 2 weeks ago

LICENSE Initial commit last month

README.md Update team information and academic term last month

README MIT license

NUBTK_COPILOT

AI-Powered University ERP and Career Architect

Course Code: CSE 4204 | Course Title: Mobile Computing Lab

University: Northern University of Business and Technology, Khulna

About

AI-Powered University ERP and Career Architect framework designed for technical skill tracking, predictive student performance profiling, and continuous automated academic career portfolio management.

Readme

MIT license

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

Your branches							
Branch		Updated	Check status	Behind	Ahead	Pull request	
oishee_database		12 hours ago		19	2		...
tisha_frontend		yesterday		19	5		...
bashir_core_backend		last week		13	1		...
dev		2 weeks ago		2	0	#10	...
ratul_ui_frontend		2 weeks ago		19	2		...

Active branches							
Branch		Updated	Check status	Behind	Ahead	Pull request	
oishee_database		12 hours ago		19	2		...
tisha_frontend		yesterday		19	5		...
bashir_core_backend		last week		13	1		...
dev		2 weeks ago		2	0	#10	...
ratul_ui_frontend		2 weeks ago		19	2		...